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Geopolitical and economic interests in environmental governance: explaining observer state status in the Arctic Council

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Abstract:
What explains the acceptance of states as members of the Arctic Council (AC) with Observer status? In 2013, the AC underwent its most significant institutionalization change since its foundation in 1996, with the creation of a secretariat. The acceptance of Observer states reflects Arctic states' goals of promoting and enhancing both geopolitical and economic interests in a context of growing concern about climate change. This study builds on the literature on intergovernmental organizations (IGOs) and environmental governance, and the factors that drive membership of IGOs. Based on linear and probit regression models, we identify that states that mostly increase their carbon-equivalent emissions through both consumption and production are less likely to join the AC as Observers. The same models also yield statistically significant correlations between states that import a high value of goods from China and Observer status in the AC. In addition, the alignment between a given country and China at the United Nations General Assembly is found to have a negative but meaningful impact on the likelihood of joining the AC as an Observer. We interpret the regression results as being the consequence of domestic pressures on governments to reiterate their commitment with climate change mitigation, and to hold Beijing accountable for its emissions. AC membership thus reflects not only concerns about climate change and sustainable development but also domestic pressures and geopolitical factors that nevertheless shape and reflect the dynamic in the two aforementioned areas (climate change mitigation and sustainable development). The choices and preferences in regard to welcoming Observer states in environmentally-focused IGOs are of pivotal importance for understanding the future of cooperation in relation to tackling global warming and its consequences.
Keywords: Arctic Council; Climate Change; Institutional Design; Intergovernmental organizations (IGOs); Environmental Regional Governance

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1 Introduction

Global conflicts put the international order under strain, putting pressure on international institutions to respond to various political and economic shifts, and to adjust in response to them. The world economy and global politics have suffered meaningful changes since Russia invaded Ukraine in February 2022. The socio-political and economic consequences of this shocking change are currently at the center of academic attention and will be investigated for years to come. The repercussions of the Russian invasion are already noticeable in international politics, including in the governance of the Arctic region. Indeed, the dynamic of power inside the Arctic Council (AC) has changed over the past ten years, as that intergovernmental organization (IGO) has pursued further institutionalization while being more selective in accepting Observer member states.

While there are those that claim that the status of Arctic states is quite obvious, as all full AC members since the council's inception in 1996 are nations with territory within the Arctic Circle, the requirements for becoming an Observer member state within the AC remain a puzzle that has yet to be solved. By employing logit and probit regression models and qualitative analysis of unsuccessful bids for acquiring Observer membership, we find that the AC welcomes states with non-full member status for environmental and geopolitical reasons. All sovereign states in the world are considered in the sample. The only exceptions are the Arctic states (Canada, Denmark, Finland, Iceland, Norway, Russia, Sweden, and the US). Our focal point of analysis is 2013, the year after the creation of the AC secretariat, and the year in which four Asian states—including China—were accepted as Observers, of which there were 12 such states at that time, most of them located in Europe. Only Switzerland was accepted as an Observer after that year, even though at least six other states have either submitted formal applications or held domestic discussions on pursuing Observer status in the AC.

We find that the likelihood of becoming an Observer state in the AC is reduced for sovereign nations that increased their carbon-equivalent emissions in both consumption and production between 1992 and 2012. In turn, the larger the value of goods imported from China, the higher the chance that in 2013 a country would be accepted as an Observer. We interpret both results as an expression of domestic pressures regarding concerns with climate change that are addressed by pursuing AC Observer status membership, as a twofold purpose is thereby achieved. First, a state that is already committed to reducing emissions can demonstrate further commitment to tackling climate change. Second, as China itself was accepted as an Observer in 2013, acquiring such status in the AC may signal to domestic audiences a willingness to hold the largest rising power in current times accountable for its contributions to global warming and for its actions to mitigate its climate change impact. In international terms, the models also indicate that there is a significant and negative correlation between being aligned with Beijing at the United Nations General Assembly (UNGA) and obtaining Observer member status at the AC.

Our findings not only contribute to the literature on IGO membership and climate change but also have policy implications considering that the AC has been under Russia's chairmanship for two years, since May 2021. As a reaction to Russia´s aggression in Ukraine, seven out of eight member states of the AC refused to attend future meetings

under Moscow's chairmanship. With the limited resumption of its work in June 2022, the AC's current functioning status can be metaphorically described as that of a sleeping bear, i.e. with uncertainty as to the way to move forward.

Given the current crossroads and crisis the AC finds itself in, it is therefore critical to re-evaluate the underlying reasons that have led to changes in the AC's institutional design, and their outcomes in terms of its institutional adaptation—toward becoming more inclusive in terms of assigning Observer status for new members.

After the AC's establishment in 1996, the number of Observer states grew from four countries in 1998 to 13 countries in 2017 (Arctic Council 2019). Ultimately, the identification of factors that explain the AC's institutional adaptation will contribute to knowledge about potential trajectories for the future institutional development of the AC in the context of the rapid climatic and geopolitical changes currently taking place.

For almost 30 years the AC has fostered international cooperation and the input of Arctic indigenous peoples aimed at enhancing knowledge to address sustainable development and environmental protection for the four million people residing across the Arctic region and for the entire global community tackling the impacts of climate change. Therefore, at a first glance, one might take for granted that the reason why most requests to become an Observer state were denied after 2013 is related to the lack of a clear reason by the applicants. In fact, after that year, only Switzerland was accepted in 2017--and only after being rejected in 2015, the same year in which Turkey presented an unsuccessful bid that had no clear justification (Limon 2021, 10).[1] Yet the fact that most prospective Observers referred to their interest in sustainable development along with their applications, and debates on the topic, suggests that further work is needed when analyzing AC Observer state membership patterns.

The growing literature on the topic focuses on understanding the economic criteria for being granted Observer states (Krasulina et al. 2019), the political dimensions of the decision, and the institutional integration of Asian states (primarily China) as Observers to the AC and China´s strategy in launching regional organizations (Agostinis and Urdinez 2021; Stephen and Stephen 2020; Ingimundarson 2014). The bulk of the literature also examines the future forms of involvement of Observers in the AC after new Observers were accepted in 2013 (Knecht 2016; Graczyk 2012). The academic discussions therefore primarily evolve around the economic and political power of Observers and their potential influence on the AC's functioning (funding and implementation of scientific projects), as well as on the composition of actors in Arctic governance as regards defining the future of the region. In this paper, we shift the research focus to assess the potential impact of countries' domestic characteristics related to climate change to understand why some applicants for Observer status in the AC are accepted and others are not.

The present study is structured as follows. We first describe the evolution of the AC from its inception until the 2013 institutionalization turn, when a secretariat was established. The following section then analyzes the theoretical literature, building on rational design theory and discussions regarding IGO membership, including debates on the universal- versus club-structure adopted by organization. Based on the gaps we identify in general, as well as in regard to understanding the case of Observers in the AC

[1] After 2013, several IGOs (the International Maritime Organization, the International Council for the Exploration of the Sea, the OSPAR Commission, the World Meteorological Organization, and the West Nordic Council) and NGOs (OCEANA) were granted Observer status within the AC. However, the reasons for their acceptance fall outside the scope of this paper.

in particular, we outline hypotheses that are tested through regression analysis and corroborated by case studies of unsuccessful applications after 2013. The conclusion outlines a potential research agenda for improving our understanding of AC dynamics, in particular its rules of membership and how they reflect not only the issue of climate change, but also great power disputes as regards consolidating and expanding worldwide influence.

2 Evolution of the AC and the Role of Observer Members

The AC was created by the Declaration on the Establishment of the Arctic Council (Ottawa Declaration) and the joint communiqué by the Arctic states in 1996. The AC is a successor of the Arctic Environmental Protection Strategy (AEPS), which was signed up to by Arctic states in 1991. The Arctic states decided that the primary focus of the AEPS would be on "deepening scientific understanding of sources, pathways, and effects of pollution in the Arctic, as well as assessing continuously threats to the Arctic environment" (Śmieszek and Koivurova 2017, 17; also see Lavelle 2021). Being based on the AEPS's principles, the Ottawa Declaration broadened the AC's mandate to include focusing on environmental protection and sustainable development.

The Ottawa Declaration categorized three groups of AC participants: members, Permanent Participants, and Observers. More specifically, the AC members were the eight Arctic states. The Ottawa Declaration and the AC Rules of Procedure stipulate that ministers of Arctic states hold the right to make decisions based on consensus concerning the AC's functioning, working, and cooperative programs, as well as financing and the admission of Observers and Permanent Participants (Arctic Council 1996). Six indigenous organizations were granted a Permanent Participant status in the AC. However, this status is not equal to that of member states since it lacks a decision-making function.[2] Specifically for Observers, the Ottawa Declaration defined this membership category as existing for "non-Arctic states; intergovernmental, inter-parliamentary, and non-governmental organizations that the Council determines to contribute to its work" (Arctic Council 1996, 3).

The 2010s started with high stakes for the AC. In May 2011, the Nuuk Declaration represented a compromise by the AC's full member states on fortifying the AC's internal structure "…and its role in responding to rapidly changing Arctic conditions..." (Saksina 2011, 3). Although the agreement that was signed focused on air and maritime search, its binding character heralded a de facto reactivation of the council and its adaptation to the challenges that climate change poses to the Arctic. Moreover, the Nuuk Meeting was a major watershed for the AC as the member states decided to create a permanent secretariat (Kankaanpää and Young 2012, 1).

Sweden took over the chairmanship of the council for two years after the meeting in Nuuk, which was also evidence of the growing interest in the Arctic outside the AC member states (Lind 2011, 6). The EU had its application rejected in 2008 before the 2011 attempt (Wallis 2011, 8). Officially the rejection of EU application stemmed from

[2] The consultative functions of the Permanent Participants grants to indigenous organizations the right to be present and raise concerns during IGO discussions and negotiations on matters of their concern. This consultative role of Permanent Participants in the AC is a result of the negotiation stage of the AEPS and AC's institutional design. Yet, in spite of recognizing the role of indigenous peoples, some Arctic states defended a restrictive role for indigenous organizations within the IGO (see Gamble and Shadian 2017). For a detailed analysis on why IGOs accept non-state members like the AC does, please see Tallberg et al. (2013) and Whytock (2005).

Brussels`s ban on seal-related products (Wallis 2011, 8). Yet during the Nuuk Meeting, member states failed to reach consensus not just on the EU's application as an Observer, but also on the requests to be admitted with the same status that were submitted by China, Italy, Japan, and South Korea. As Bergh (2011, 21) argued in that context, there was seen to be a risk that the AC would be regarded as a club, "...taking major decisions about the Arctic with little regard for the concerns and interests of non-Arctic states."

Applications by those four states are reported to have been submitted by 2007. Due to the lack of clear rules on accepting or rejecting applications for Observer status and their role once they join the AC, a decision was taken only in 2013. China had demonstrated an interest in the Arctic at least since 1999, when Beijing first organized an expedition by sea to the region, something that happened twice again, in 2003 and 2008 (Jakobson 2011, 24), before its application for Observer status was submitted. It was in this context, that China started to reflect on policies to maximize its benefits in terms of trade and energy matters, as climate change had made it easier to navigate the Arctic, at least during summer, thus opening a new commercial route between East Asia and Europe (Jakobson 2011, 25-26). Specifically, since 2012 China has claimed to be a near-Arctic state as developments in the circumpolar region directly affect the weather and, hence, the agriculture and the economy of China (Stephen and Stephen, 2020, 56).

The year 2013 was therefore a watershed for the AC and may be characterized as the high point of a general change of the organization. According to Thompson (2010, 270), such a change provides the flexibility needed for an international institution to be transformed in such a way that all of its members can address new changes. In turn, a particularistic change allows certain actors to diverge from the existing rules, while the institution itself remains unchanged (Thompson 2010, 270-271). Those that currently hold the category of Observers within the AC include 13 non-Arctic states, 13 intergovernmental and inter-parliamentary organizations, and 12 NGOs.[3] After initial mistrust, the EU finally acquired ad-hoc Observer status in the AC, which in practice is similar to the official Observer-level membership (Hossain 2015).

From 2013 onwards, however, the requirements for becoming an Observer became more detailed, with the costs of membership increasing. The revised Arctic Council Rules of Procedure further specified the criteria for Observer status: among other parameters, these include recognition of the sovereignty and sovereign rights of the Arctic states, respect for the culture and traditions of Arctic indigenous peoples, and willingness and financial ability to contribute to the work of Permanent Participants and the work of the AC (Arctic Council 2013a). The primary role of Observers is to observe the work of the AC and to contribute to its work at the level of its Working Groups (Arctic Council 2013b). The Observer Manual defined Observers' consultation capacities during discussions and negotiations within the AC, stating that they may present statements and documents only at the discretion of the Chair (Arctic Council 2013b). Observers do not have voting rights. Finally, the AC also defined Observers' financial contributions to its projects, stating that these should not exceed the fees paid by member states (Arctic Council 2013b).

In 2015, during a ministerial meeting, further amendments were introduced to the Observer Manual that redefined the initial restrictions imposed that limited Observers' influence in the AC (Graczyk and Koivurova 2014). One of the provisions of the revised Manual prescribes that "the detailed information should be sent to the Senior Arctic Officials (SAOs) if any subsidiary body wishes to accept the offer from Observers to host

[3] For a full list of Observers, see Arctic Council (2022).

a meeting or to receive funding larger than those of the eight Arctic states" (Knecht 2016, 603). The ramifications of these amendments include the enlarged scope of actors that can make a financial contribution to the work of the AC. This might have a positive effect in the future on an increase in the number of projects and activities implemented by the AC. Ultimately, the introduced amendments preserve the prerogative position of the Arctic states in decision-making regarding financial support for the AC's activities.

In regard to this article, what remains puzzling is why, even after the redefinition of Observers' attributions, it seems that there is no consistency as to which states are accepted as new members with Observer status. That is why we now turn to revisiting the literature on the rational design of international institutions and works on IGO membership. Both types of work have the potential to shed light on the continuous demand for Observer status within the AC.

3 Theorizing IGO Membership

The rational design theory considers "institutions as rational, negotiated responses to the problems international actors face" (Koremenos et al. 2001, 786). In this case, states create institutions to promote their interests and develop a structural design of international institutions accordingly, which includes membership rules (Koremenos et al. 2001). Such a design, in turn, would be a function of the following factors: distribution problems; enforcement problems; the number of actors and asymmetries among them; and uncertainty about state behavior and preferences, as well as about the state of the world (Koremenos et al. 2001).

Other works based on rational design theory aimed either to test the original framework or to fill in the gaps by unpacking the proposed dimensions and variables. Specifically related to observer membership status within IGOs, it is worth noting that states may prefer to join IGOs and to sign up to agreements that offer escape clauses that allow members to refrain from full commitment to institutional obligations. Such escape clauses—among which we include membership as an observer only—contribute to addressing domestic uncertainty and flexibility (Rosendorff and Milner 2001).

Yet uncertainty and flexibility also arise at the international level. Within IGOs, flexibility allows states to make robust commitments and to reach durable agreements (Koremenos 2009). Uncertainty, in turn, often arises when leading members of organizations define issue areas and the tasks of the organization (Wang 2015). Arguably, the AC faces a high degree of uncertainty due to the changing dynamic of its main issue area (the Arctic and related environmental issues that have relevant geopolitical implications, since the world's two main nuclear powers—Russia and the US—are AC members).

Also, becoming an observer may be attractive in organizations with a low degree of delegation, which tends to be the case for task-specific international institutions that often pool authority through majoritarian decision-making (Lenz et al. 2015). Moreover, when states that do not belong to a given IGO face problems of asymmetric information vis-à-vis member-states, the costs of joining the organization tend to lead non-members to cooperate once they join the IGO (Koremenos et al. 2001, 784). Both mechanisms clearly apply to the AC, given its focus on the Arctic and environmental issues related to the region, and the production of information and knowledge on the same topics.

Arguably, the AC's institutional design resembles that of a club IGO, insofar as such a type of organization allows (at least originally) member states to focus on a specific issue area while being opaque to outsiders, as a means of achieving efficiency (Keohane and Nye 2001). With a restricted number of members, there is a trend toward like-

mindedness (Drezner 2003) through convergence around the problems to be solved and eventually the solutions themselves. For instance, in the origins of the multilateral trading system and its development in the form of the General Agreement on Trade and Tariffs (GATT) in the pre-World Trade Organization (WTO) from 1948 to 1995, there was club due to the control of the negotiating agenda by only rich countries (Keohane and Nye 2001). Likewise, as Drezner (2003, 5) claims, great powers employ club IGOs when they regulate financial issues.

Despite being partially based on constructivist elements like status, most recent scholarship corroborates the above points on club IGOs, which stem from the rational design project. In analyzing the case of the GATT/WTO, Davis and Wilf (2017) concluded that membership in the multilateral trading system was de facto contingent upon geopolitical considerations and voting affinity at the UNGA. These findings are aligned with the argument that not only do clubs provide exclusive information, they are also a mean for selling status (Cornes and Sandler 1996), which is an element that IGOs can provide to states that are affiliated to them (Gray 2013).

In sum, there might be pre-conditions for accepting a state as a member (even with observer status only) attached to its domestic and international characteristics, including alliances with other sovereign nations and their political regime (Mansfield and Pevehouse 2006, 2009). Indeed, based on the case of the Organization for Economic Co-operation and Development (OECD)—often dubbed "the club of the rich"—Davis (2016) claims that IGO membership is not random.

In addition to status reasons, which impact the demand side of IGO membership, the supply side also matters insofar as member states tend to act as gatekeepers in order to preserve key characteristics of the club/IGO. In the same vein, powerful states may prefer retaining control over membership rules (or lack thereof) as a means of shaping the IGO according to their interests (Stone, Slantchev and London, 2008; Stone, 2011). Yet those states may be trumped by weaker ones, which can engage in blackmailing to extract concessions from the most powerful members in exchange for agreeing to accept newcomers in IGOs that require unanimity in decision-making (Schneider and Urpelainen 2012).

In addition, the supply side of IGO membership is also impacted by the pursuit of legitimacy. While clubs tend to be efficient, as explained above (Drezner 2003), due to the likelihood of converging interests, IGOs that open to universal membership may see their decisions and roles mostly accepted by the international community in general—including non-members. This seems to have been the trend since the end of the Cold War and the rise of globalization because power dispersion between and within states facilitates network diplomacy against the club, exclusive model (Heine 2013).

Therefore, openness to NGOs and non-state actors in general can increase accountability, and hence an IGO's legitimacy. The AC has pursued this route since its onset, as detailed above. Considering this, is the acceptance of Observer states within the AC also related to the pursuit of legitimacy? The AC may just be following the trend identified by Greenhill and Lupu (2017) for IGOs: although they tend to have regionally-based membership, "...states are increasingly working through IGOs with states outside their own clusters" (Greenhill and Lupu 2017, 193). This possibly stems from the fact that "[i]f an institutional arrangement restricts the benefits of cooperation to members, actors have an incentive to pay the price of admission to the club" (Koremenos et al 2001, 783). However, given the restrictions on the roles played by members with Observer status in the AC, acceding to the organization has relatively few benefits in terms of reducing information asymmetries.

Hence, we can conclude there is still a puzzle as regards comprehending the major reasons behind the Arctic states' decisions to grant Observer status to sovereign entities outside the region. One strategy for moving this debate forward lies on the conclusions of Stephen and Stephen (2020). Based on the case of the AC's expansion in 2013, Stephen and Stephen (2020) developed a framework for analyzing situations in which clubs accept emerging powers. First, these powers will be accepted insofar as they are perceived as fitting into the club. Second, on the one hand, the club must offer goods that make membership worthwhile (a supply-side element), and, on the other, the emerging powers should demonstrate that the IGO receives gains by accepting them as members. Third, acceptance of new members is a form of institutional adaptation that opens the doors to the use of integrative strategies instead of distributive ones. In sum, membership of the AC has yielded more gains since the onset of the twenty-first century than before as it emerged as a focal point for establishing new policy, discussing issues, and producing and disseminating scientific knowledge related to a region that is increasingly affected by the consequences of climate change (Stephen and Stephen, 2020).

Stephen and Stephen's framework is applied to the case of China only. By the beginning of the twenty-first century this rising state was the fastest growing economy in the world and Beijing foresaw gains from joining the AC in terms of reducing information asymmetries on the Arctic, and hence the effects of climate change. In turn, at a first glance the AC's full members were persuaded of the advantages of China having Observer status, and of the gains for individual council members. However, the fact that, apart from Beijing, regional opponents to China, like India, Singapore, and South Korea, were also accepted as Observers in 2013 suggests that not only rising powers but also emerging markets can have enough status to enter IGOs. We here apply Gomez Bruera's (2015) four-criteria to distinguish between rising powers from emerging markets: to be classified as the latter, it does not suffice having economic growth and/or a large internal market. A rising power needs to contest (albeit softly) the Western-centric international status quo, apart from being a regional leader that nevertheless has a well-defined international identity and the ability of influencing the global order. For instance, in being an EU member, Estonia can only be considered an emerging market, largely thanks to its comparative advantage in information technology. It had its application for Observer status rejected in 2021, whereas two rising powers—Brazil and Turkey, which do not automatically align themselves with the West and are regional powers in South America and the Middle East respectively—showed interest throughout the 2010s in obtaining the same status.

3.1. *Outlining the Testable Hypothesis*

The above discussion indicates that there is space, beyond explanations based on information asymmetries, for further scrutiny of the reasons why states are eventually granted Observer status membership in the AC. However, rather than focusing on states' status and great power interests only, we argue that we must assess the potential impact of countries' domestic characteristics related to climate change to understand the acceptance of applications for Observer status in the AC. Domestic audience costs may lead governments, regardless of their political regime—democratic or autocratic—to demonstrate competence and resolve on climate change. Joining IGOs that address the challenge of climate change may be a relatively costless way to meet those demands. We then posit the following hypotheses:

- Hypothesis 1a: The higher the variations in CO_2-equivalent emissions in consumption, the lower the chance of a state joining the AC as an Observer.
- Hypothesis 1b: The higher the variations in CO_2-equivalent emissions in production, the lower the chance of a state joining the AC as an Observer.

In an economically globalized world, increasing consumption-equivalent emissions inevitably relate to imports. This market process generates negative externalities for the environment. As has been widely documented in the literature on the topic, the migration of manufacturing from the advanced industrial democracies to Eastern Asia, in particular, has been linked to rising levels of carbon emissions. Despite having higher levels of economic welfare, consumers may react against environmental destruction by targeting large polluters, regardless of the destination of their production. We then consider the following hypothesis:

- Hypothesis 2: Countries that import from China may face domestic pressures to tackle global warming by holding Beijing accountable for its emissions.

Joining the AC satisfies this goal. For the council member states—in particular, Western states—accepting more Observers may help check Chinese ambitions in the Arctic. Beijing's opponents may be more willing to play the role of holding China's authorities accountable for excessive emissions. We then consider the following hypothesis:

- Hypothesis 3: Countries that are mostly opposed to China in ideological terms have a higher chance of being accepted as Observers in the AC.

4 Empirical Analysis

A key methodological challenge for assessing the reasons why a state is accepted as an Observer to the AC stems from, as discussed above, the lack of clear rules on granting a state such a status. For instance, in 2015, Turkey applied for Observer status for the first time. According to Limon (2021), there is no document that explains why it was rejected. Canada and Russia reportedly oppose the expansion of the number of Observers (Tømmerbakke 2019). The unsuccessful Swiss application was submitted under Ottawa's Chairmanship of the IGO (2013–2015), whereas the last official applications by Czechia, Estonia, Ireland, and Turkey, during the beginning of Russia's chairmanship, were unsuccessful. Even before the security crisis 2022, Russia was not acknowledged as not being particularly pro-environmentally inclined at governmental level (see Kochtcheeva 2021: Hartwell et. al. 2021). Yet, participation in regional organizations, such as the AC; but also multiple others and reference to environmental agenda was almost routinely (Lavelle 2021; Hartwell 2021; Agostinis and Urdinez 2021). This chairmanship is still officially in place (2021–2023), even though other AC member states have refused to meet as a group under Russian leadership, in reaction to Moscow's invasion of Ukraine.

Therefore, we first rely on regression models to test the hypotheses outlined above. In the light of the results, we provide an overview of unsuccessful bids and we conclude this section with a discussion of the data triangulation between the quantitative and qualitative results.

4.1 Data

Our dependent variable is *Observer status* and is binary, being coded as 1 if in 2013 a given country was an Observer at the AC and 0 otherwise (Arctic Council 2022). All

member states (Canada, Denmark, Finland, Iceland Norway, Russia, Sweden, and the US) are excluded, by default, from the sample. China is also excluded, given that we test hypotheses related to the growing political-economic impact of that power in the process of climate change and the expansion of the number of state Observers accepted by the AC. Otherwise noticed, all independent variables and controls are lagged in one year.

Two variables test hypothesis 1 and its variants a and b. Change in CO2 Emissions in Consumption (1992–2012) tests hypothesis 1a, whereas Change in CO2 Emissions in Production (1992–2012) tests hypothesis 1b (Our World in Data 2022). For both we consider the period beginning in the first year after the collapse of the Soviet Union (1992), which heralded the beginning of the consolidation of the globalization agenda in non-economic issues, such as environmental change, democracy, and human rights. In addition, for environmental/climate change issues, the year 1992 was a watershed, with the United Nations Summit in Rio de Janeiro, Brazil, after which multilateralism for tackling global warming in particular expanded, beginning with the creation of the Global Environmental Facility and leading to rising awareness about the climate question among non-state actors all over the world.

As discussed above, the environmental question is, however, also related to the concentration of industrial production in East Asia—more specifically in China. This may generate domestic pressure for states to act internationally against global warming and to pressure Beijing on this issue area through IGOs, such as the AC, which has accepted China as an Observer. We then test whether high values of imports from China correlate with Observer status in the AC (Observatory of Economic Complexity 2022). Given the dispersion of the observations, we logged the variable, which is originally measured in current USD.

On the supply side, as summarized above, member states may have wanted to attract and retain Observers that could limit China's clout within the IGO. We then operationalize hypothesis 3 by predicting a negative correlation between Observer status and alignment with China at the UNGA (Voeten et al. 2009). Table 1 summarizes the variables we work with.

Apart from the variables that operationalize our hypotheses, we employ three controls. First, it is plausible to consider that democracies tend to be more open than autocracies to domestic civil society demands related to climate change. Moreover, the literature claims that democracy makes states more prone to joining IGOs (Mansfield and Pevehouse, 2006, 2009) To control for these potential effects, we employ the V-Dem Polyarchy Index (Coppedge et. al 2021). Observer status may be a luxury that only the most powerful states can afford. Given that the US and Russia are excluded from the sample as members, and China is also excluded, as the Observer which is of primary interest for testing our hypotheses, we predict that the composite index of national capabilities (CINC) has no significant effects (Singer et al. 1972). This is also the case for GDP per capita (World Bank 2022), as emissions equivalent may matter more than wealth for triggering domestic concern on the environment. We thus predict that any effect of a state's relative wealth will be irrelevant for determining Observer status. Both variables were lagged due to the large dispersion of observations.

Table 1. Descriptive Statistics

Variable	**Obs**	**Mean**	**Std. Dev.**	**Min**	**Max**
Observer Status	185	0.064865	0.246956	0	1
Change in CO_2 Emissions in Consumption (1992–2012)	109	0.598899	0.947761	-0.56	4.54
Change in CO_2 Emissions in Production (1992–2012)	183	0.75929	4.417361	-0.76	59.32
Imports from China (log)	183	20.83221	2.30936	13.78758	25.81731
Alignment with China at UNGA	184	0.8262	0.10953	0.331395	1
V-dem Polyarchy Index	164	516.7256	261.9432	9	913
CINC (log)	179	-2.95011	2.50274	-9.21034	3.092519
GDP per Capita (log)	181	8.583619	1.405304	5.530857	11.96731
European State	185	0.205405	0.405094	0	1

Finally, we include a dummy to account for states located in Europe (Singer et al. 1972), whose countries are close to the Arctic and have material conditions on average which allow them to bear the costs of joining the AC. In addition, the dummy controls for states that historically have had higher status than the average within the international community (Watson 1992). Even those that were not colonial metropolises achieved sovereign status on average earlier than Global South states.

4.2 Research Design and Results

Given that the dependent variable is binary, we run logit and probit models, alternating between models with robust clusters according to European status and regressions with uncorrected standard errors. The models with robust standard errors consider de facto fixed effects stemming from a state's location (i.e. in Europe or not).

Overall, we find strong support for hypothesis 1a (Table 2) and hypothesis 2. Yet the variable "Alignment with China at UNGA" has only statistically significant effects in the models with clustered robust standard errors (models 1, 3, table 2; models 5, 7, table 3), thus yielding mixed conclusions for hypothesis 3. The same applies for hypothesis 1b, whose proxy "Change in CO2 Emissions in Production (1992–2012)" lacks significance in models 6 and 8 (Table 3). The difference may stem from the fact that the sample for testing hypothesis 1b has more observations than the one applied to hypothesis 1a because of limitations in the availability of data on consumption-related emissions.

Table 2. Does Variation in Emissions Related to Consumption Correlate with Observer Status in the Arctic Council?

VARIABLES	Model 1 Logit	Model 2 Logit	Model 3 Probit	Model 4 Probit
Change in CO2 Emissions in Consumption (1992–2012)	-9.283**	-9.283*	-5.226**	-5.226*
	(4.115)	(5.407)	(2.062)	(2.922)
Imports from China (log)	6.358***	6.358*	3.651****	3.651*
	(2.094)	(3.814)	(1.043)	(2.141)
Alignment with China at UNGA	-3.854***	-3.854	-2.708****	-2.708
	(1.462)	(12.52)	(0.462)	(7.219)
V-Dem Polyarchy Index	0.00667	0.00667	0.00381	0.00381
	(0.00794)	(0.00623)	(0.00444)	(0.00353)
CINC (log)	0.00614	0.00614	-0.0251	-0.0251
	(0.634)	(2.068)	(0.441)	(1.228)
GDP (log)	-1.985*	-1.985	-1.124**	-1.124
	(1.014)	(1.768)	(0.545)	(1.037)
Constant	-134.6***	-134.6*	-77.12***	-77.12*
	(46.06)	(78.12)	(22.63)	(43.23)
Robust-Cluster for Europe	YES	NO	YES	NO
Observations	103	103	103	103

****p.<0.001 ***p.<0.01 **p.<0.05 *p.<0.1

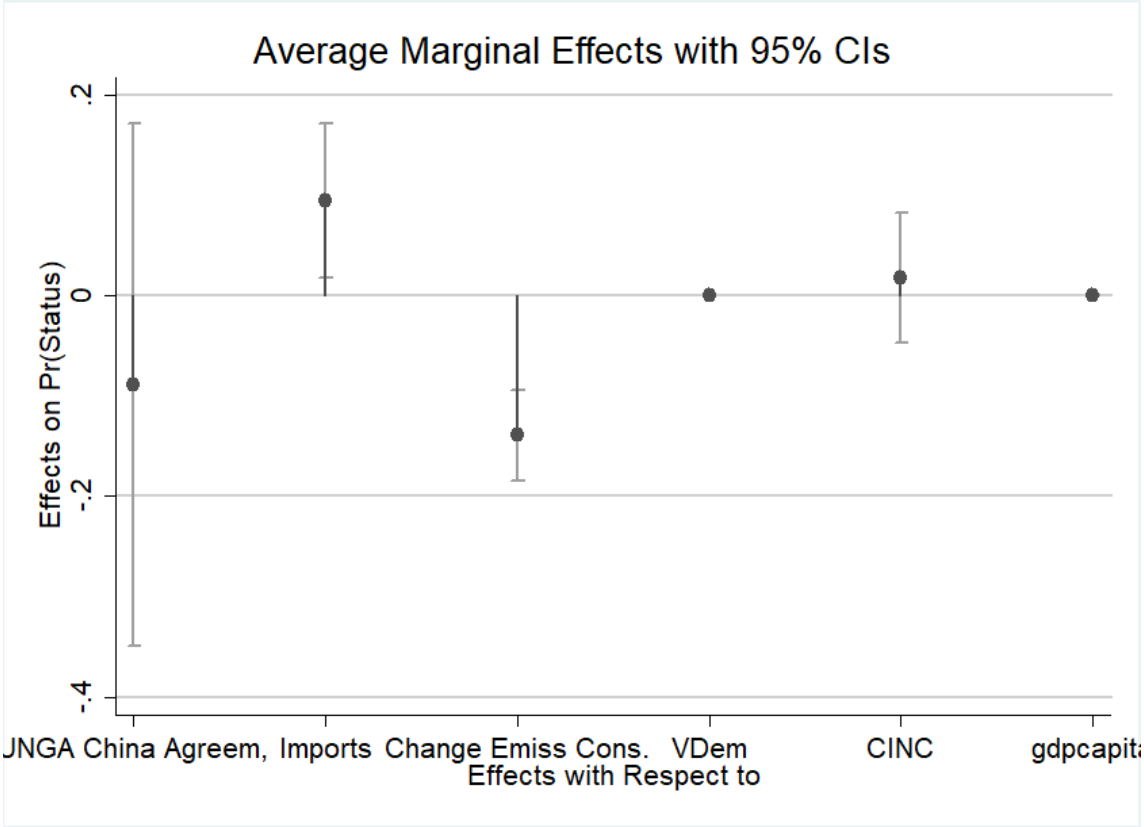
Table 3. Does Variation in Emissions Related to Production Correlate with Observer Status in the Arctic Council?

VARIABLES	Model 5 Logit	Model 6 Logit	Model 7 Probit	Model 8 Probit
Change in CO2 Emissions in Production (1992–2012)	-5.448** (2.716)	-5.448 (3.341)	-3.044** (1.329)	-3.044 (1.891)
Imports from China (log)	4.975*** (1.793)	4.975* (2.934)	2.821**** (0.642)	2.821* (1.674)
Alignment with China at UNGA	-7.341**** (0.182)	-7.341 (10.53)	-4.410*** (1.381)	-4.410 (6.006)
V-Dem Polyarchy Index	0.00471 (0.00830)	0.00471 (0.00567)	0.00254 (0.00456)	0.00254 (0.00325)
CINC (log)	-0.0574 (0.619)	-0.0574 (1.670)	-0.0143 (0.214)	-0.0143 (1.007)
GDP (log)	-1.660* (0.928)	-1.660 (1.634)	-0.933** (0.375)	-0.933 (0.956)
Constant	-100.9*** (39.13)	-100.9* (58.07)	-57.09**** (15.58)	-57.09* (33.00)
Robust-Cluster for Europe	YES	NO	YES	NO
Observations	156	156	156	156

****p.<0.001 ***p.<0.01 **p.<0.05 *p.<0.1

The interpretation of non-linear models depends on the marginal effects of each variable. Figure 1 plots the impact of each variable for model 1 from table 2. Imports from China (log) increase the chances of obtaining Observer status by almost 10%, whereas changes in consumption-equivalent CO2 emissions decrease the probability of acquiring Observer status by just less than 14%.

Figure 1. Marginal Effects of Independent Variables



The statistical results are corroborated by the cases of six states that either officially pursued Observer status or at least demonstrated the intention of doing so. In 2015, Switzerland applied to become an Observer. However, at the Iqaluit Ministerial Meeting at that time, no decision was made. Then, in the subsequent meeting in Fairbanks (Alaska, USA), in 2017, that country was finally accepted. The 2021 AC ministerial meeting reported four applications for Observer status (Czechia, Estonia, Ireland, and Turkey). However, the meeting's final declaration did not mention any decision on this matter. Brazil reportedly signaled is desire to become an Observer twice: once in 2012, when its BRICS partners China and India were on the way to reaching that status, and another in 2019. Yet no official application has yet been made.

4.3 Understanding Unsuccessful Bids

The goal of this section is to understand why some bids are successful and others are unsuccessful, by assessing the qualitative evidence provided by both primary and secondary sources related to applicants for Observer status in the AC after 2013. We cover the case of Switzerland, which presented an unsuccessful bid in 2015 and was finally accepted in 2017, Czechia, Estonia, Ireland, and Turkey, all of which last attempted to accede to the IGO in 2021, and Brazil, which has never submitted a formal candidacy but which is a helpful case for understanding how eventual AC Observer membership can be a means either of attaining higher international status or just reaffirming a state's position as main global power.

Environmentally Related Explanations beyond Emissions: Switzerland claimed membership on the grounds of being a "vertical Arctic nation" (Arctic Council 2017), a clear reference to the alpine characteristics of its territory. Czechia's arguments for

obtaining Observer status resembled Switzerland's: although Prague did not claim to be a "vertical Arctic nation", it did refer to the existence of an Arctic-Alpine tundra biome in its territory (Ministry of Foreign Affairs 2021). Yet, unlike the Swiss bid, Czechia's one was rejected.

In addition to the rejection of Czechia's bid, Estonia's and Ireland's unsuccessful applications also furnish evidence of the fact that it does not suffice to be affected by Arctic-related climate change to be welcomed as an AC Observer state. For instance, Ireland's bid was rejected even though its coastal and farming communities are affected by environmental transformation in the Arctic (Quinn 2021).

In addition, the Czech authorities emphasized the country's experience in researching the Arctic (Lanteigne 2021). Estonia's and Turkey's arguments also included this factor, which, as detailed below, may also reflect economic concerns.

Science and Technology: A Missing Link with the Economic Realm? Turkey has invested in scientific studies related to the Arctic since the beginning of the twenty-first century (Limon 2021). Likewise, as summarized in the Estonian government webpage on its application to the AC as an Observer, "Estonia is the northernmost non-Arctic country. We are directly affected by developments in the Arctic. With its knowledge, long-time experience in polar research, and inventive approach, Estonia has a lot to offer to the sustainable development of the Arctic. This is why Estonia is applying for Observer status on the Arctic Council" (Ministry of Foreign Affairs 2021).

The structure and length of internal discussions in Estonia also illustrate the issue-linkage that exists between science and technology, on the one hand, and economic matters, on the other. The country submitted its application in November 2020, to be considered at the AC Ministerial Meeting of May 2021, in which Russia took over the chairmanship from Iceland. Intragovernmental discussions on the topic, along with civil society consultations, had started a year before the submission (Ministry of Foreign Affairs 2021). The application was formally rejected due to the lack of consensus among member states (EER News 2021). Shortly after, the Estonian government announced that it would target the 2023 AC Ministerial Meeting (Embassy of Estonia Oslo 2021), thus indicating its hope that Estonia might follow Swiss's fate.

The case of Brazil—which has never submitted a formal application—also illustrates how complex are the domestic scientific-technological-economic motivations that lead a state to become interested in membership of the AC. In 2019—seven years after the first news of Brazilian interest in the AC emerged (Correio Braziliense 2012) — a colloquium organized by the Navy of Brazil resumed the discussion, which was also supported by the Ministry of Foreign Affairs. According to the ministry, Brazil had an interest in research on climate change, partnerships in the field of oil and gas, information on shifts related to global logistics—a clear reference to the expanding conditions of navigation in the Arctic Seas (Santos et al. 2018)—and knowledge of mining under the seafloor (Tømmerbakke 2019). Moreover, it was argued that Brazilian experience and technology in offshore oil drilling could be of interest to the AC's members for exploring natural resources in the Arctic, as discussed in that colloquium, with the support of a Norwegian consultancy firm (Hellebust 2019). In turn, it was suggested that Brazil could expand its knowledge of the state of the art on climate change and sea protection—a major issue for Brazilian strategic interests in the South Atlantic (Hellebust 2019; Santos et al. 2018).

The Pursuit of Status and Geopolitical Considerations: Like Brazil's intentions with regard to joining the AC, more than expressing scientific concerns, Turkey's pursuit of

Observer status may reflect the goal of achieving "greater political penetration on a global scale" (Limon 2021, 2). The context in which Brasilia first expressed an interest in joining the AC corroborates such an argument.

In 2012, Brazil's BRICS partners China and India were on the way to obtaining Observer status, which was confirmed a year later. Local media reported in 2012 Brasilia's goal of becoming an Observer, along with Beijing and Delhi (Correio Braziliense 2012), in clear pursuit of parity status with other so-called emerging powers. Debates on the matter had begun two years before the first application attempt was considered.

Brazil and Turkey also adopted similar tactics in their approaches to the AC. In 2018, the Government of Turkey began internal discussions on joining the Svalbard Treaty, which recognizes Norway's sovereignty over that archipelago, as a means of leveraging its likelihood of being accepted as an Observer in the AC (Limon 2021).[4] Like Ankara, Brasilia also considered joining the Svalbard Treaty (Tømmerbakke 2019).

By contrast, Estonia downplayed proximity with its Nordic neighbors like Norway—with which Brazil and Turkey were seeking proximity—and preferred to mitigate eventual Russian opposition, with American support. As already discussed, the Estonian application was submitted at the beginning of Russia's chairmanship in the AC. As in other cases, no reason for rejecting Estonia's application was given (Kirschner 2021). Yet it is worth noting that a paper published by the Heritage Foundation, a main US Conservative think tank, defended the idea that Washington should endorse Estonia's bid, based on the country's status as a reliable North Atlantic Treaty Organization (NATO) ally (Coffey and Kochis 2021).

Another piece of evidence that the Estonian government seemed to be actively pursuing US support for its obtaining Observer status in the AC is the fact that it co-sponsored, along with the Polar Institute (a Washington-based think tank focused on Arctic-related issues), a seminar discussing the application and the impact of eventual membership of the AC (Wilson Center 2021). In contrast to the military approach, which Heritage defended, the event—which was hosted at the Wilson Center (a major centrist US think tank)—focused on Estonia's potential contributions to sustainable development in the Arctic, which nevertheless remains a backdoor for meddling in geopolitical issues that may oppose great power interests in the AC.

At a first glance, the outcome of Estonia's application could therefore be a result of the Russian perception that the country would be a mere proxy of American interests in the AC. Yet, military neutrality is no guarantee that Observer status will be granted. Were that the case, Ireland—which like Estonia has geographical proximity to the Arctic and is neutral like Switzerland (which has Observer status)—would not have had its application dismissed in 2021.

4.4 Discussion

The analysis of the scores of the applicant states within the variables that yielded meaningful results, from the regression analysis, clarify the reasons why those nations were unsuccessful in their bids to obtain Observer status (Table 4). Estonia's denied membership demonstrates the limits of the findings, as it has continuously reduced its emissions. Turkey, in turn, is aligned with China at the UNGA, which partially explains the lack of consensus among AC members on accepting Ankara as an Observer. Yet the

[4] For a discussion of Turkey's potential benefits from joining the treaty, see Çetin and Büyüksağnak (2021).

score of Turkish alignment with Beijing is below the average. Only Brazil's alignment with China has a score above Turkey's. Such a fact reduces the chances that Brazil becomes an Observer. By contrast, the Brazilian import-dependence vis-à-vis the Chinese, with more than USD 30 billion in purchases (the highest of all cases) in 2012, could increase domestic pressures for holding Beijing accountable for emissions and, hence, the demand for joining IGOs such as the AC in which such a goal could be accomplished.

Table 4. States Interested in Becoming AC Observers after 2013

Country	Change in CO2 Emissions in Consumption (1992–2012)	Below Average ?	Imports from China (USD Billion)	Above Average ?	Alignment with China at UNGA	Below Average ?
Brazil	0.65	NO	36.42	YES	0.88	NO
Czechia	-0.26	YES	11.04	NO	0.67	YES
Estonia	-0.11	YES	1.41	NO	0.68	YES
Ireland	-0.17	YES	2.63	NO	0.72	YES
Turkey	0.20	YES	19.45	YES	0.76	YES
Mean	0.59		11.9		0.82	

Brazil therefore turns out to be the state that is least likely to be granted Observer status. Turkey, by contrast, is the only prospective Observer that scored in all variables according to the regression results. Hence, one could expect that it would be likely to become the next state to be accepted as an AC member with Observer status. Yet, considering that—like Estonia—Turkey is a NATO member, it may be an ongoing victim of Russia's assertiveness since Russia's invasion of Crimea in 2014. An important conclusion for future studies is therefore that the likelihood of obtaining Observer status in the AC increases with increased uncertainty about non-members' preferences and behavior. New members could check not only China's positions on climate change and geopolitics but also Russia's stance on the same matters.

5 Conclusion

The AC was established as an intergovernmental forum that was small in scale, without a secretariat or a budget but in the period 2011–2013 it underwent structural changes, with the creation of a secretariat, an increasing number of Observer participants, and a widening of the scope of issues with which it engages, thus acquiring the characteristics of a de facto IGO.

This article has presented quantitative and qualitative evidence for assessing what explains the acceptance of states with Observer status in the AC. The results of logit and probit models corroborate the hypotheses we outlined. As predicted, the higher the variations in CO_2-equivalent emissions in consumption, the lower the chance of a state joining the AC as an Observer (Hypothesis 1a). Likewise, the higher the variations in CO2-equivalent emissions in production, the lower the chance of a state joining the AC as an Observer (Hypothesis 1b). In addition, countries that import from China may face domestic pressures to tackle global warming by holding Beijing accountable for its

emissions (Hypothesis 2) and Observer status in the AC can be directed towards such a goal. Finally, countries that are mostly opposed to China in ideological terms have a higher chance of being accepted as Observers in the AC (hypothesis 3) due to the fact that all permanent members but Russia are Western and, hence, tend to oppose Beijing's views on political matters expressed at the UNGA.

In the light of Russia's invasion of Ukraine and Moscow's tensions with the EU and the US, the unsuccessful cases of Observer status application indicate that likelihood of obtaining restricted membership in the AC increases with a high degree of uncertainty about members' preferences and behavior. In this vein, Russia is very likely to keep vetoing Turkey as it is a NATO member. Likewise, other Arctic states (all of which are Western) may oppose an eventual Brazilian application as Brasilia is aligned with Beijing at the UNGA. Both conclusions are crucial also considering that the potential increase in financial support provided by Observers to the AC's projects might create a situation in which they will want to redefine their status in the decision-making process in future.

Future works could explore whether a reason for the eight Arctic states to support the admission of new Observers has been existing budget constraints. More specifically, the budget limitations hinder the AC's capacity to address promptly and effectively the impacts of climate change in the region. The acceptance of new Observers might contribute to the growth of the AC's scientific projects since new non-members contribute with financial allocations and expertise to the work of the AC, which is in line with the literature which sees IGOs as a means of pooling resources (Abbott and Snidal 1998; Lenz et al. 2015). In this way, this finding reflects the existing literature that argues that IGOs are interested in new members' resources and knowledge, which can allow them to tackle problems by bringing in expertise and financial allocations (Tallberg et al. 2013).

In addition, future research on the transformation of the AC's institutional setup must account for dimensions other than membership. For instance, the expansion of the scope of issues engaged with by the AC has come as a reaction to the emerging climatic and socio-economic challenges. The changes in the range of issues engaged with have led to the institutional transformation of the AC, including the establishment of the secretariat and task forces to address the emerging challenges. yet it remains unclear how exactly the expansion of issue-areas happened.

Finally, future research must also scrutinize patterns that might explain the acceptance of NGOs as Observers. The number of NGOs with Observer status expanded from six in 1998 to 26 in 2019 (Arctic Council 2019). Unlike Observer states, NGOs can serve as reliable allies for permanent, Arctic states to legitimize consensus and eventually sort out disputes with each other in climate change-related topics within a controlled institutional setting such as the AC.

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